

Pokhara University
Faculty of Management Studies

Course Code: CMP 174

Course Title: Digital Systems

Nature of the Course: Theory & Practical

Level: Bachelor

Credits: 3[Hrs]

Total Lectures: 48 hours

Program: BCSIT

Year: I Semester: II

1. Course Description

This course offers a comprehensive introduction to digital systems and logic design, covering key topics such as number systems, Boolean algebra, logic gates, and the simplification of Boolean functions. It includes practical design and implementation of both combinational and sequential circuits, focusing on essential components like adders, subtractors, decoders, encoders, multiplexers, demultiplexers, and code converters. Students gain in-depth knowledge of flip-flops, shift registers, and counters, crucial for sequential logic. The course culminates in the design of core digital components, including the Arithmetic Logic Unit (ALU), accumulator, status register, and processor unit. This course provides a balanced mix of theoretical knowledge and hands-on experience, equipping students with the skills necessary to understand, design, and analyze complex digital systems.

2. General Objectives

The general objectives of this course are:

- To equip the students with the fundamental concept of logic system.
- To provide students with basic tools for the design of digital circuits.
- To provide students a strong grasp of computer hardware basics and software for problem-solving, utilize accessible learning, hands-on activities, and examples.

3. Specific Objectives and Contents

Specific Objectives	Contents
• Define of number systems • Conversion between decimal, binary, octal, and hexadecimal systems for both integer and floating-point values. • Determine of 1's and 2's complements of binary numbers. • Execute the arithmetic operations on binary numbers	Unit I: Binary Foundation and Digital Representation (8 hrs) 1.1 Introduction to Digital System 1.2 Binary Number System Hexadecimal and Octal Number Systems 1.3 Number system conversion 1.4 Binary Codes: weighted and non-weighted codes 1.5 Alphanumeric codes (ASCII, EBCDIC) 1.6 Representation of Negative Numbers 1.7 Subtraction using complements
• Recognize the laws and theorems of Boolean algebra • Comprehension of logic gates, truth tables. and their practical implementation • Proficiency in implementing universal gates in digital circuitry	Unit II: Boolean Building Blocks (6 hrs) 2.1 Basic Laws of Boolean Algebra 2.2 De Morgan's Theorems 2.3 Logic Gates and Their Symbols 2.4 Logic Gate Truth Tables 2.5 Universal Gates
• Understand the concepts of the	Unit III: Simplification of Boolean Functions (6

minterms and maxterms in Boolean algebra. • Develop the proficiency in simplifying Boolean functions using sum of products and product of sums methods. • Gain proficiency in determining the dual form and complement of Boolean functions. • Improve the proficiency in employing Karnaugh maps and handling don't-care conditions to streamline Boolean function simplification.	hrs.) 3.1 Minterms and Maxterms 3.2 Sum of products and Product of sums 3.3 Dual and complement of Boolean function 3.4 K-map & don't care conditions 3.5 Two-Level and Multilevel gate Implementations
• Develop design methodologies for combinational logic circuits • Become skilled in creating adders and subtractors • Master in decoder and encoder circuits, along with their practical uses • Become Capable in designing and employing multiplexers and demultiplexers • Become Proficient in various code conversion methods like 8421, BCD, Gray, Excess-3, and 2421 • Learn the principles and applications of Programmable Logic Devices	Unit IV: Combinational Logic (9 hrs.) 4.1 Introduction to combinational Logic circuit 4.2 Adders and Subtractors 4.3 Decoder and Encoder 4.4 Multiplexer and Demultiplexer 4.5 Code Conversion (8421, BCD, Gray, Excess-3, 2421) 4.6 Programmable Logic Device (PROM)
• Understand the functionalities and uses of diverse types of sequential logic circuits • Master the activation processes of flip-flops • Learn to handling different flip-flop variants like RS, JK, T, D, and master-slave. • Develop in creating state diagrams and tables to represent sequential logic systems.	Unit V: Sequential Logic (6 hrs.) 5.1 Types of sequential logic circuit 5.2 Triggering of flip flops 5.3 Flip-flops (RS, JK, T, D, master-slave) 5.4 State Diagrams and State Tables
• Learn how shift registers work and their different modes (SISO, SIPO, PISO, PISO) • Develop proficiency in designing and implementing asynchronous counters	Unit VI: Registers and Counters (6 hrs) 6.1 Shift Register 6.2 Modes of shift register (SISO, SIPO, PISO, PISO) 6.3 Asynchronous counter (Binary ripple, CD

such as Binary ripple and BCD counters • Improve the skill in designing and operating synchronous counters, capable of both up and down counting	counter) 6.4 Synchronous Counter (up/down counter)
• Become Proficient in designing a 4-bit Arithmetic Logic Unit (ALU) • Improve the skill of accumulator design and operation. • Understand the functionality of status registers and flags • Escalate the competence in designing a processor unit for digital systems.	Unit VII: Digital Systems Design (7 hrs.) 7.1 Arithmetic Logic Unit (Design of 4-bit ALU) 7.2 Accumulator 7.3 Status register and flags 7.4 Processor Unit

Note: The figures in the parentheses indicate the approximate periods for the respective units.

4. Laboratory Works

1. Implement AND, OR, and NOT gates using basic electronic components.
2. Verify truth tables for each gate through practical experimentation.
3. Implement the simplified Boolean expressions using logic gates.
4. Verify De-Morgan's theorem
5. Design of half adder & full adder
6. Encoder and decoder
7. Multiplexer and Demultiplexer
8. Design of flip flops
9. Design of shift register and counters

Experiential activities can be showcased through hands-on implementation on a digital logic trainer kit or simulated using dedicated software.

5. Methods of Instruction

- Lecture
- Group discussion
- Question-answers
- Demonstration and discussion
- Presentations
- Guest lectures
- Group work/project work
- Problem solving
- Simulation
- Tutorials

6. Evaluation system and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student's Responsibilities

Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and

References Text Books

1. Morris Mano (2017). Digital logic and computer design. Pearson India

References

1. Fletcher, J. R. (2017). Digital Logic and Computer Design. Wiley
2. Mano, M. M., & Ciletti, M. D. (2021). Digital Design: With an Introduction to the Verilog HDL. Pearson
3. Malvino, A. P., & Leach, D. P. (2010). Digital Principles and Applications. McGraw-Hill Education